mijnDUO
D4F Assignment

Use case:

Scenario:
After three years of hard work, Maria finished her bachelor of Information Sciences. Now that she has finally received her bull at the graduation ceremony, she is ready to apply for a master. She wants to apply in time for the master Information Sciences as there are only limited places available. In order to register for this master, she has to login to mijnDUO which is an online environment from the Dutch government focused on services concerning educational institutes. In order to login to mijnDUO, Maria has to login to DigiD, which is a system which allows you to identify yourself when making arrangements on the internet, such as with mijnDUO. After finally being able to login (Maria always has problems with her DigiD) she discovers that her high school diploma, bachelor's diploma and grade lists are all missing in her online environment. Maria cannot add those documents manually, however in order to apply for the master, her past diplomas and grade lists should all be uploaded. She does not know where to find the correct information, and how to access it. Furthermore, the data provided in mijnDUO is not interchangeable and reusable. Maria needs to apply for her master as soon as possible, what to do?

Tasks & needs
Maria needs to:

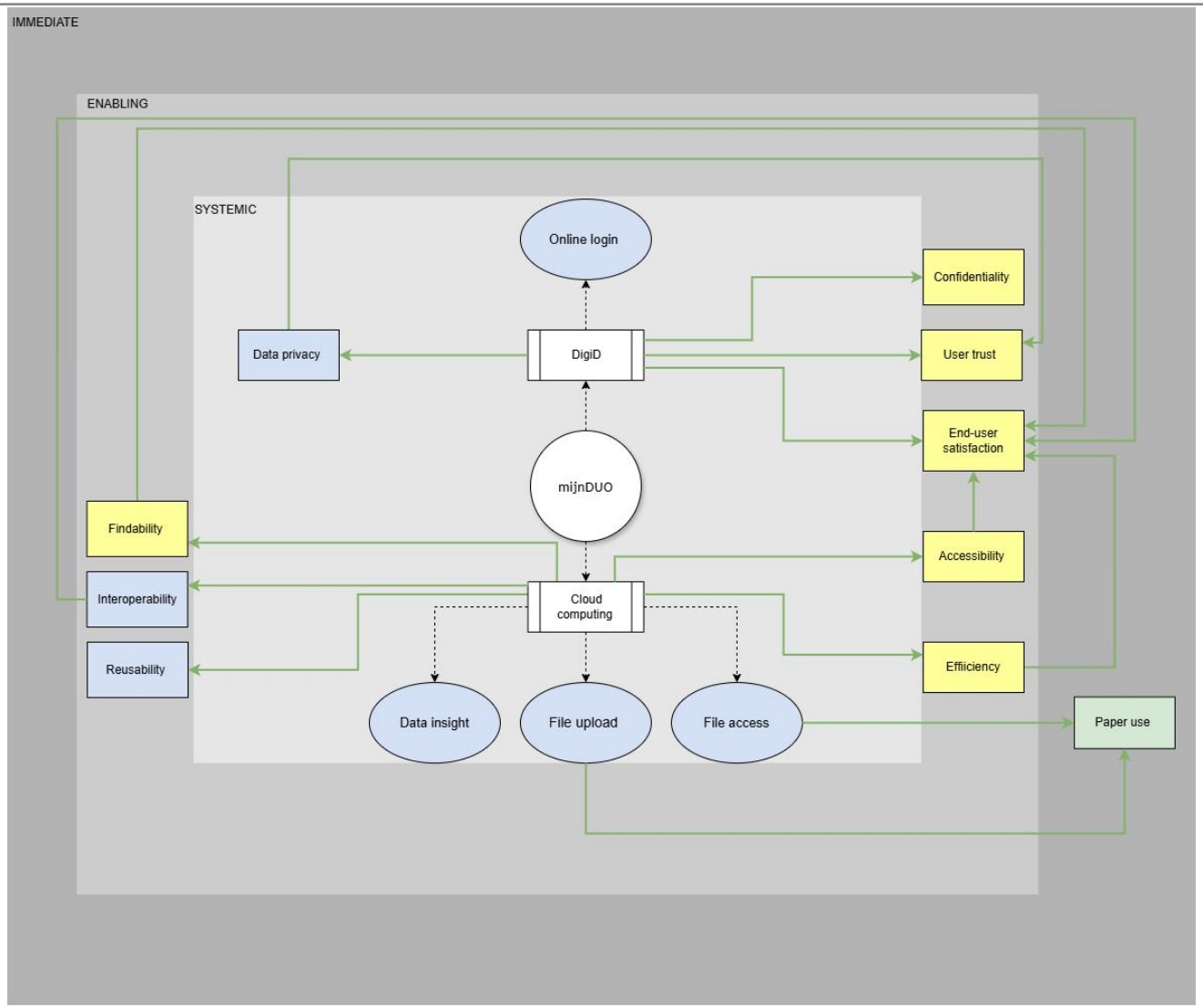
- Login to mijnDUO
- Gain insight into her data
- Access important files
- Upload important files
- Submit files for master's registration

Environment: mijnDUO platform, this is a platform concerning educational institutes in the Netherlands. The information is stored in a cloud-environment.

Challenges

- Maria does not know where to find and how to access the correct information
- Maria has never given consent for mijnDUO to store all her personal data
- mijnDUO has not registered Maria's past diplomas and grade lists, and now she cannot upload it herself
- The data in mijnDUO is not interchangeable and reusable for other platforms

Definition of Fairness: Decision Map



Definition of Fairness: Decision Map (cont.)

The mijnDUO system is dependent on two technological features, namely DigiD and cloud computing. DigiD is an authenticator from the Dutch government which allows you to securely login online to Dutch governmental websites. For one, DigiD ensures confidentiality, which means that the data is protected against unauthorized access. It also results in a higher user trust, user trust the resource (mijnDUO) more. Furthermore, it ensures that the end-user satisfaction is higher, the end-users are more satisfied with the resource (mijnDUO). And lastly, it makes sure that there is a high data privacy, which means that the end-users have control over their own personal data.

Cloud computing is the availability of computing resources on demand. As mijnDUO uses cloud computing, users can have insight into their data, can upload files and access files on demand. The cloud computing thus makes accessing files, uploading files and gaining insight in data more efficient; the end-goal of using the resource (mijnDUO) is reached in the best way possible. It also makes the resources more accessible, the data and the resource are stored in such a way that they are easily accessible for both humans and other computer systems.[1] Furthermore, it makes sure the resource and data is better findable. Which entails that the metadata and the resource are easily found by both humans and computer systems.[1] The use of cloud computing also results in a better interoperability and reusability. The data can both easily be exchanged and interpreted by humans and other computer systems, and the data and metadata are described in such a way that it can be reused in the future.[1] The heightened efficiency, accessibility (including usability, inclusivity and operability) and findability also result in a higher end-user satisfaction, as the resource is easier to use as a whole.

Accessibility is linked to usability, inclusivity and operability, these quality attributes are all different aspects of accessibility and help enhance this attribute, these are explained in the SQ model. Lastly, the function to upload and access files results in less paper use, which is better for the environment. Because the cloud computing and DigiD ensure a higher findability, reusability, interoperability, confidentiality, end-user satisfaction, data privacy, user trust, accessibility and efficiency, we believe that the system is more fair. The system and the data become easier to use for its end-users, having less troubles using the system, which we believe makes it more fair.

The table below describes each element of the decision map.

mijnDUO	Online environment from the Dutch government focused on services concerning educational institutes.
DigiD	An authenticator from the Dutch government which allows you to securely login to Dutch governmental websites.
Online login	The feature to be able login online to mijnDUO.
Cloud computing	Is the availability of computing resources (on-demand).
Data insight	The possibility/ feature to look in personal data which is stored in mijnDUO.
File upload	The possibility/ feature to upload files onto mijnDUO, files like diplomas, grade lists, etc.
File access	The possibility/ feature to access files in mijnDUO, files like diplomas, grade lists, etc.
Paper use	The amount of paper used for printed files.
Reusability	Data and metadata are well described so it can be reused in the future. [1]
Interoperability	Data can be exchanged and interpreted by humans and other computer systems. [1]
Findability	The metadata and the resource are easily found by humans and computer systems. [1]
Confidentiality	The protection of the data against unauthorized access.
Data privacy	The way the end-user has control over their own personal data.
End-user satisfaction	The way end-users are satisfied with the resource.
User trust	The way users trust the resource.
Accessibility	The resources and the data are stored in such a way that they are easily accessible for both humans and computer systems. [1]
Efficiency	Attaining the end-goal of using the resource in the best way possible.

1. Open Access and Fair Principles

Definition of Fairness: SQ model

QUALITY ATTRIBUTE (QA)	DEFINITION	REFERENCE	TECHNICAL	ENVIRONMENTAL	ECONOMIC	SOCIAL	DEFINITION of METRIC/KPI
Interoperability	degree to which mijnDUO can exchange information with other systems and use the information	simplified from [ISO/IEC 25010:2011]	X				percentage of correctly performed data exchanges
Efficiency	resources expended in relation to the accuracy and completeness with which users achieve goals	[ISO/IEC 25010:2011]				X	average time for correctly using the resource (login/data insight/file upload/file access)
Reusability	degree to which an asset can be used in more than one system, or in building other assets	[ISO/IEC 25010:2011]	X				percentage of correctly reused datasets by other systems
Accessibility	degree to which mijnDUO can be used by people with the widest range of characteristics and capabilities	simplified from [ISO/IEC 25010:2011]				X	percentage of users having accessibility problems
User trust	degree to which a user or other stakeholder has confidence that mijnDUO will behave as intended	[ISO/IEC 25010:2011]				X	percentage of users complaining about the system
Confidentiality	degree to which mijnDUO ensures that data are accessible only to those authorized to have access	[ISO/IEC 25010:2011]				X	percentage of unauthorized users/ systems accessing unauthorized data
Data Privacy	the degree to which the collection, storage, or use of personally identifiable information is protected	adapted from [Condori-Fernandez et al. (2020)]	X				number of privacy related user complaints
Paper use	amount of paper used for documents	adapted from Yale Sustainability		X			amount of files uploaded/accessed online instead of printing it
End-user satisfaction	degree to which end-users are satisfied with mijnDUO	adapted from [ISO/IEC 25002:2024]				X	average user satisfaction rate
Findability	degree to which mijnDUO and metadata are found by the end-users and other computer systems	adapted from Interaction Design Foundation				X	percentage of successful data access
Usability	degree to which a system can be used by specified users to achieve specified goals	[ISO/IEC 25010:2011]				X	percentage of people who complete tasks correctly
Inclusivity	degree to which a system accommodates diverse user needs and characteristics	adapted from [ISO/IEC 25010:2011]				X	percentage of success criteria met at WCAG
Operability	degree to which a product or system has attributes that make it easy to operate and control	[ISO/IEC 25010:2011]	X				time it takes for users to complete common operations

Definition of Fairness: SQ model (cont.)

The SQ model shows the most important Quality Attributes in order to help assess and improve a software system, in this case the mijnDUO system. In order to make the mijnDUO system fair for all possible end-users, ten quality attributes have been defined, as well as the corresponding metrics/KPIs. The ten quality attributes that have been defined in the SQ model correspond with the concerns in the decision model. We believe that for mijnDUO to become fair for each and every end-user, the system should be interoperable, efficient, accessible, confidential, and findable. We also believe that the user trust should be of a high level, as well as the data privacy and end-user satisfaction. Furthermore, we believe that the paper use should be reduced to make it fair for the environment as well. The metrics/KPIs in the last column explain how each quality attribute can be measured. This way, it is possible to measure whether the system continuously meets our fairness requirements.

By designing the system in line with these requirements and by monitoring them according to the measures/KPIs, we will ensure that Maria can complete her tasks and needs. She will be able to successfully login to mijnDUO, gain quick and easy insight into her data, easily access and upload important files and submit the required files for her master's registration. Furthermore, it will help her in tackling her challenges. If the system is accessible and findable, she will easily know where to find and how to access the correct information. Furthermore, the confidentiality and data privacy will ensure that Maria feels safe in giving consent to mijnDUO for storing all her personal data. Whenever the system becomes more efficient as well, Maria will be able to upload the necessary files herself. However, due to the interoperability and reusability, mijnDUO will also become more efficient in registering her past diplomas and grade lists itself, taking the task away from Maria and enhancing the efficiency. And lastly, due to the interoperability and reusability the data in mijnDUO will become interchangeable and reusable for other platforms as well. The implementation of these quality attributes will not only highthen the end-user satisfaction and user trust, it will also reduce paper use which is an environmental plus.

Incorporating these quality attributes and measuring them, will thus ensure that mijnDUO becomes a more fair system for all its end-users.

Fairness perspective

	Description
Desired quality	Fairness for all users by ensuring the user satisfaction, accessibility of the system, privacy of data, security of the system. A fair system should meet the expectations of the user, always be accessible for the user, it has to make sure the private data from users is kept safe and the system is protected with crucial security elements.
Concerns	Concerns representing the desired quality are: user satisfaction and user trust, meeting the expectations of the user and providing consistency. Accessibility, meaning equal accessing for all users, supporting access from several devices, no language barriers and a user-friendly interface. Data privacy, protecting all personal data, transparency in data collection and users being able to control personal data. Finally, security, preventing unauthorized access and security against possible data breaches.
Architectural tactic(s)	The ensure fairness for all user of the system, the system has to be accessible for all users. Easy access with a friendly user-interface and helpful guidelines can help users access and upload necessary documents when needed [1].
Possible KPI(s)	Percentage of potential end-users with perfect access to mijnDUO **= f(accessibility, findability, efficiency, security)**
Problems and pitfalls	Security might ask for extra authentication which could reduce satisfaction because of the extra operation and extra time it takes for a user to login. Besides this, complex authentication might make it less accessible for users with limited digital experience. If the system is easily available for everyone and manual uploading of documents is the systems efficiency and performance might be compromised.

[1]: **World Wide Web Consortium (W3C).** (2008). *Web Content Accessibility Guidelines (WCAG) 2.0* (B. Caldwell, M. Cooper, L. G. Reid, & G. Vanderheiden, Eds.). https://www.w3.org/TR/WCAG20/

Definition of KPI*

Sustainable access to data in mijnDUO					
Goal	**Critical Success Factor (CSF)**	**Key Performance Indicator (KPI)**		**Metric(s)**	**Measure(s)**
Make mijnDUO accessible, findable, secure and efficient for each and every potential end-user	By 2035, ensure that mijnDUO is accessible, findable, secure and efficient for each and every potential end-user	Percentage of potential end-users with perfect access to mijnDUO **= f(accessibility, findability, efficiency, security)**		**Accessibility** = f(login success, navigation success rate) **Findability** = f(search success rate, navigation success rate, structure of data and metadata) **Efficiency** = f(ease of use, response time) [2] **Security** = f(authentication level, user trust rate, data protection level, number of reported security issues) [3]	- Login success - Navigation success rate - Search success rate - Navigation success rate - Structure of data and metadata - Ease of use - Response time - Authentication level - User trust rate - Data protection level - Total end-users - Reported security issues
		Target	**Action(s)**		
		Percentage of citizens with perfect access to mijnDUO should be **above 90%**	Improve the software structures of mijnDUO Integrate a working cloudcomputing environment		

[2] Albertao, F., Xiao, J., Tian, C., Lu, Y., Zhang, K. Q., & Liu, C. (2010, November). Measuring the sustainability performance of software projects. In *2010 IEEE 7th International Conference on E-Business Engineering* (pp. 369-373). IEEE.
[3] Mishra, S., Vi, P. T., Phuc, V. M., Oni, D., & Tanh, N. V. (2023). Using security metrics to determine security program effectiveness. *Human Factors in Cybersecurity*, *91*(91).

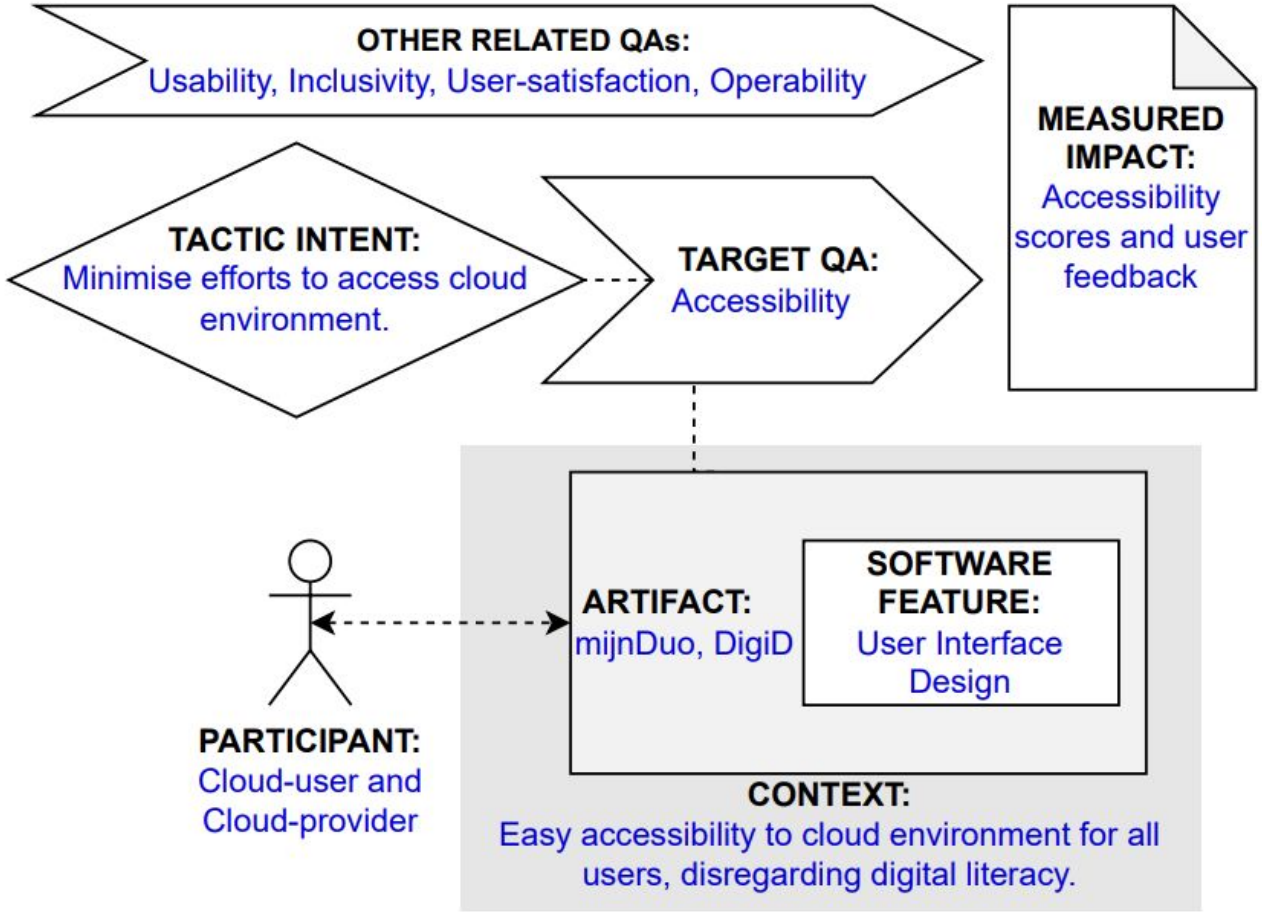
Architectural Tactic(s) for Fairness*

Attribute	**Description**
Title	Accessibility in Cloud-Based Applications
Description	Accessibility of personal documents through online government services often depend on the cloud provider's platform. The cloud-based interfaces should be accessible and usable for all users, disregarding their digital literacy. This means easy accessibility and navigation for online government services. Available additional guidelines and friendly interfaces to increase the accessibility of the platform.
Participant	Cloud-user and cloud-provider
Related Artifact	mijnDuo, DigiD
Context	Normal operation
Feature	Accessible user interface including clear navigation, guidelines for assistance and a consistent layout. Serving users with different abilities and varying levels of digital literacy.
Target QA	Accessibility
Related QA's	Usability, inclusivity, user satisfaction, operability

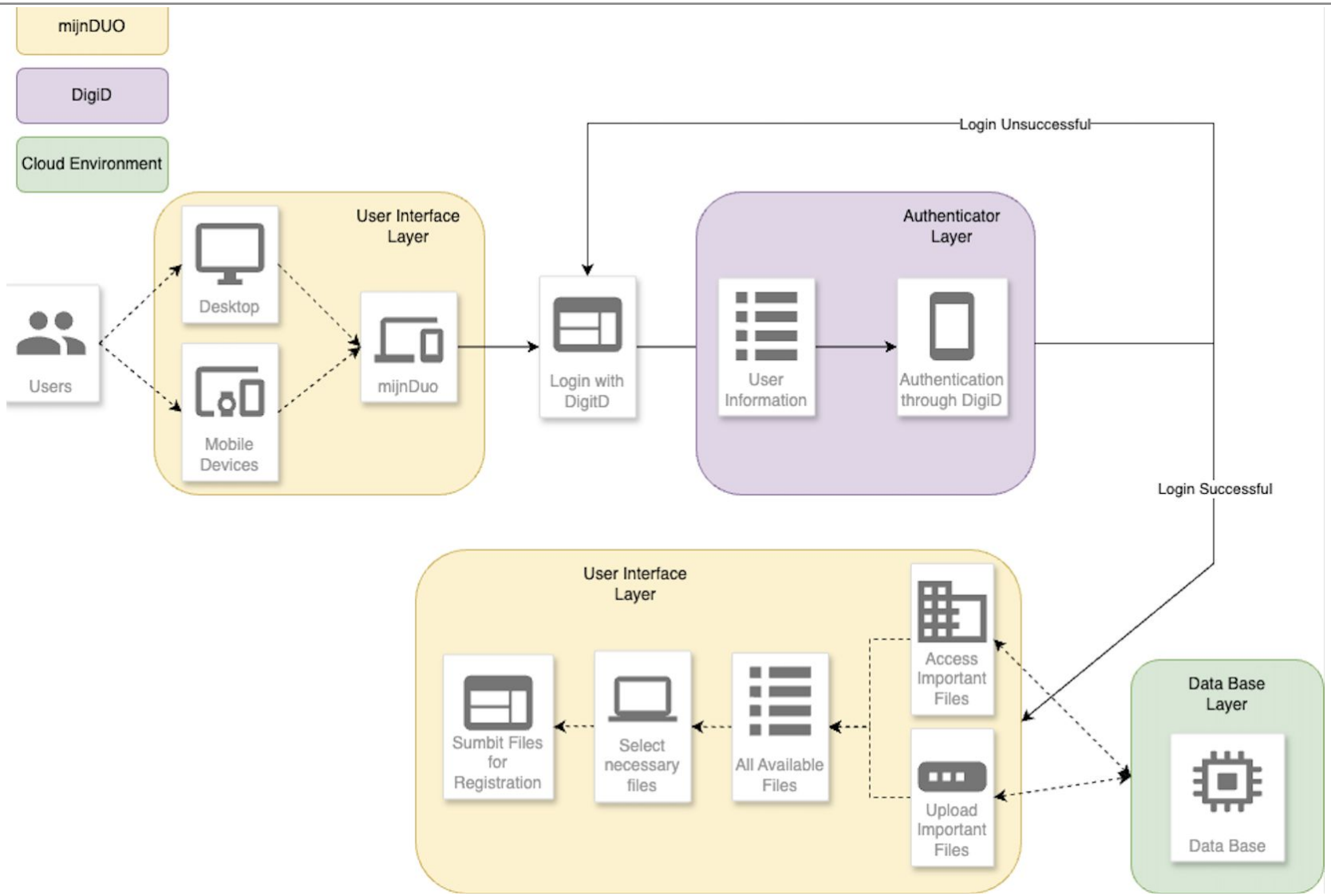
Architectural Tactic(s) for Fairness*

Attribute	**Description**
Tactic Intent	Ensure access to cloud-environment for all users by minimizing efforts required to interact with the system and improving guidance for the online service.
Intent Measure	Accessibility audit scores (WCAG 2.0 compliance), success rates from users, feedback from users.
Counter Measure	Test the accessibility features across different devices to ensure user satisfaction.
Source	World Wide Web Consortium (W3C). (2008). *Web Content Accessibility Guidelines (WCAG) 2.0*. https://www.w3.org/TR/WCAG20/

Architectural Tactic(s) for Fairness*



Fairness view: mijnDUO for everyone



Fairness view: mijnDUO for everyone

The Fairness view for the accessibility of mijnDuo presented on the previous page will be explained here, in relation to the user case of Maria. First of all maria wants to login using her desktop or any other mobile device. Using one of the devices she enters the online government environment where she can navigate easily towards to login button. When she wants to login she has to confirm her identity using the DigiD authentication. Her personal information and DigiD codes are needed to confirm her identity so she can safely enter her online government environment. After authentication, the login is either successful or unsuccessful. Unsuccessful attempts will be send back to the initial login page, requiring the user to start the identification process from the start. A successful login will bring the user to the user interface including their personal information. The user will be able to access existing files or upload new files manually, after which a full list of available files is presented. From these available files Maria can select the ones she needs for her registration for her Master. After registration she can safely log-out of her online government environment. With this view Maria can easily access her online document, the documents are either already available or she can upload them manually. This increases the accessibility of the platform. The use of the cloud environment enhances the interoperability and the reusability of the system. Furthermore, the accessibility is enhanced by the fact that the environment can be accessed through multiple devices and contains a clear user interface with guidelines for navigation. Using Digid for login increases the privacy and security for the user. All together these quality attributes increase the user satisfaction which is important for keeping the online environment accessible for all users.